

A Computer-Assisted Proof of the Oriented Ramsey Value $k(3, 4) = 21$

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

Let $k(3, 4)$ be the least positive integer N such that every loopless oriented graph on N vertices contains an independent triple or a complete transitive tournament on four vertices. We give a computer-assisted proof that $k(3, 4) = 21$. A previously recorded cyclic graph supplies the lower bound. For the upper bound, elementary neighbourhood restrictions and an odd-order parity argument reduce a hypothetical graph on 21 vertices to 129 template cases. Eight checked classification refutations establish coverage of the local templates. A further 108 checked refutations, together with 21 explicit reversal transports, exclude every case. Acceptance of the certificates relies on a stricter LRAT checker: an independent computational audit found that the original deposited checker could accept an unsupported empty clause and replayed the entire selected corpus after correcting that defect. The proof retains an ordinary software and compiler trust boundary; it is not a proof-assistant formalization. We make no priority claim over an earlier assertion of the same value.

1 Statement and context

An oriented graph is a finite directed graph without loops in which each unordered pair has either no arc or exactly one of its two possible arcs. Write I_s for the edgeless graph on s vertices and L_t for the transitive tournament on t vertices. In particular, an L_4 contains all six pairs. An acyclic four-vertex graph with a missing pair is not an L_4 . All subgraphs considered as forbidden configurations below are induced.

Definition 1. *The number $k(3, 4)$ is the minimum positive integer N such that every oriented graph with exactly N vertices contains an I_3 or an L_4 .*

This is the parameter $r(I_3, L_4)$ in the notation of Ihringer, Rajendraprasad, and Weinert [3]. Their work studies this family of oriented Ramsey numbers. Its Lemma 3.1 already establishes uniqueness, up to isomorphism, of the $\{I_3, L_3\}$ -free graph of order eight; the local certificate system here independently recovers the needed classification. That classification and the elementary degree method are not claimed as new mathematical ideas.

The working report for Erdős problem 112 [2, Section 6] records the cyclic order-20 construction used below and the interval $21 \leq k(3, 4) \leq 25$. These are historical statements about that report, not a claim that the parameter remained publicly open at the time of this manuscript. Kirtchakov's repository [4, Part G] already asserts $k(3, 4) = 21$. Its description says that the classification representatives and queue files for its upper-bound computation are private and that its LRAT corpus was deleted after verification. We have not independently validated that upper-bound argument. The present proof uses its own retained classification and refutation corpus, without relying on that

assertion or treating the missing public inputs as evidence against it. No priority or exhaustive novelty claim is made.

For nearby context, an ordered-split-graph result determines a different Ramsey invariant, the small Ramsey degree of an edge [6]. For the certificate methodology, one-sided parity compression has been used to construct refutation-sound SAT encodings with a checked DRAT-to-LRAT pipeline for an Erdős–Szekeres problem [5]. Neither work determines the oriented Ramsey number considered here.

Theorem 2. *For loopless oriented graphs, the oriented Ramsey number satisfies*

$$k(3, 4) = 21.$$

Equivalently, there exists a $\{I_3, L_4\}$ -free graph on 20 vertices, and every oriented graph on 21 vertices contains an independent triple or a complete transitive four-vertex tournament.

The proof consists of the exact lower-bound check in Section 2, the reductions in Sections 3–5, and the certified finite exclusions in Section 6. The certificate files are part of the mathematical evidence; this text does not replace their replay with an assertion about a solver’s answer. Section 7 specifies the replay and its trust boundary.

2 The lower bound

Let D have vertex set $\mathbb{Z}/20\mathbb{Z}$ and arcs

$$x \longrightarrow x + c, \quad c \in C = \{2, 4, 5, 7, 11, 14\}.$$

This is the construction recorded in [2, Section 6]. Since $0 \notin C$ and $C \cap (-C) = \emptyset$, it has neither loops nor antiparallel arcs. It has $20 \cdot 6 = 120$ arcs.

Proposition 3. *The graph D has no independent triple and no L_4 .*

Exact finite verification. Construct the arc set directly from 20 and C . For each of the $\binom{20}{3} = 1140$ triples, test whether any ordered pair is an arc. The number with no arc is zero. For each of the $\binom{20}{4} = 4845$ four-sets, first require an arc on every unordered pair. For each complete four-set, test acyclicity by repeatedly deleting a vertex of indegree zero in the remaining induced graph. The number of complete sets for which all four vertices can be deleted is zero. A finite digraph is acyclic exactly when this deletion succeeds: a nonempty acyclic digraph has a source, whereas a remaining digraph with no source contains a directed cycle, obtained by following incoming arcs. An acyclic tournament is transitive, by its topological ordering. Thus these two exact tests check precisely the forbidden configurations. The independent verifier reconstructs the arc set from the connection set and reports the stated counts. A separate deposited graph verifier uses the ordering characterization of a transitive tournament. Both checks agree. $\square$

For clarity, the essential four-set test can be reproduced by the following pseudocode; it uses no numerical approximation.

```
for Q in all four-element subsets of {0,...,19}:
  if some pair of Q has no arc in either direction: continue
  R = Q
  while R is nonempty:
    find v in R with no incoming arc from R
```

```

    if no such v exists: break
    delete v from R
assert R is nonempty

```

Proposition 3 gives $k(3, 4) > 20$. Every smaller positive order also has an avoiding graph by taking an induced subgraph of D .

3 Elementary bounds and root selection

For a vertex v , let $N^+(v)$ and $N^-(v)$ be its out- and in-neighbourhoods, and let $N^0(v)$ be its non-neighbours other than v . Write $d^+(v), d^-(v), d^0(v)$ for their sizes. These sets partition the vertices other than v .

Lemma 4. *A $\{I_3, L_3\}$ -free oriented graph has at most eight vertices. An L_4 -free tournament has at most seven vertices. Consequently, every vertex of a $\{I_3, L_4\}$ -free graph satisfies*

$$d^+(v) \leq 8, \quad d^-(v) \leq 8, \quad d^0(v) \leq 7.$$

Proof. In a graph without I_3 or L_3 , any arc inside a directed neighbourhood would complete an L_3 with its root. Both directed neighbourhoods are therefore independent and have at most two vertices. The non-neighbours form a tournament, since a missing pair together with the root would be an independent triple. Every tournament on four vertices has an L_3 : at any vertex, at least two of the other three vertices are on the same directed side, and their connecting arc completes a transitive triangle. The non-neighbourhood thus has at most three vertices. The order is at most $1 + 2 + 2 + 3 = 8$.

In an L_4 -free tournament, either directed neighbourhood avoids L_3 and has at most three vertices by the preceding tournament observation. Its order is at most $1 + 3 + 3 = 7$. Finally, in a $\{I_3, L_4\}$ -free graph, either directed neighbourhood avoids both I_3 and L_3 , and the non-neighbourhood is an L_4 -free tournament. Apply the two bounds. $\square$

In particular every avoiding graph has order at most 24. This proves finiteness of the minimum in the definition without an external bound. Equality in the order-eight local bound forces the profile $(2, 2, 3)$ at every vertex of such a local graph.

Lemma 5. *If a $\{I_3, L_4\}$ -free graph on 21 vertices exists, then, after possibly reversing all arcs, it has a root with profile (d^+, d^-, d^0) in*

$$(8, 8, 4), (8, 7, 5), (8, 6, 6), (8, 5, 7), (7, 7, 6).$$

For the last branch one may additionally assume $d^+(u), d^-(u) \leq 7$ for every vertex u .

Proof. If some directed degree is eight, global reversal makes it an outdegree and the vertex may be chosen as root. Its other two degrees sum to 12; Lemma 4 leaves precisely the first four profiles. Otherwise both directed degrees are at most seven everywhere. Since $d^0 \leq 7$, every total degree $d^+ + d^-$ is 13 or 14. All 21 total degrees cannot equal 13: their sum would be 273, but the sum of total degrees is twice the number of arcs. A vertex of total degree 14 then has profile $(7, 7, 6)$. $\square$

The last condition is a condition on the chosen branch, not a universal assertion about every root with profile $(7, 7, 6)$ in an arbitrary graph. A vertex with profile $(7, 6, 7)$ need not be impossible at this stage; the argument selects a different root when required.

4 Certified local classifications

Let a_s denote the number of directed isomorphism classes of $\{I_3, L_3\}$ -free graphs on s vertices, and b_t the number of L_4 -free tournaments on t vertices. The certified local inventories are

s	5	6	7	8	t	4	5	6	7
a_s	25	31	7	1	b_t	3	3	1	1.

Here is how completeness, rather than just production of these lists, is certified. A local graph is stored by a word on $\{0, 1, 2\}$ over the unordered pairs in increasing lexicographic order. For a pair $u < v$, these symbols mean no arc, $u \rightarrow v$, and $v \rightarrow u$, respectively.

For the directed classes, choose a root and partition the remaining vertices into its three neighbourhoods. Enumerate all profiles bounded by $(2, 2, 3)$ and summing to $s - 1$. At order eight the only profile is $(2, 2, 3)$. Each directed block is independent. The non-neighbour block is a tournament without L_3 , hence is empty, a single vertex, an arc, or a directed triangle according to its order. Relabel within that block to fix the arc or cycle orientation. Every local graph has a labelling of this form. In particular, normalization does not identify arbitrary graphs with their converses.

The coverage formula encodes the local forbidden-subgraph definition and these normalizations, using a disjunction of admissible profiles when needed. For every listed normalized record it adds the clause excluding exactly that directed-arc assignment. Unsatisfiability proves that no normalized graph was omitted. A separate backtracking isomorphism verifier matches every record to a listed representative, verifies that each representative occurs, and distinguishes the representatives pairwise. For tournaments, the coverage formula uses the unrestricted tournament definition and excludes every relabelling of the listed representatives. Its unsatisfiability directly establishes coverage without relying on the tournament producer's normalization.

There are four directed and four tournament coverage refutations. Their clauses are reconstructed from the definitions by a separate implementation and compared as multisets with the deposited formulas. The eight LRAT refutations all pass the stricter replay described in Section 7. Enumeration code and previously printed class counts alone are not used as completeness certificates.

Apply these inventories to the three blocks of a root in Lemma 5. The directed blocks have a_{d+} and a_{d-} possibilities, and the third has b_{d0} possibilities. Relabelling the blocks independently fixes their templates and leaves every cross-block pair unrestricted. The resulting cover is

Root profile	Template cases	Direct refutations	Reversal transports
$(8, 8, 4)$	$1 \cdot 1 \cdot 3 = 3$	3	0
$(8, 7, 5)$	$1 \cdot 7 \cdot 3 = 21$	21	0
$(8, 6, 6)$	$1 \cdot 31 \cdot 1 = 31$	31	0
$(8, 5, 7)$	$1 \cdot 25 \cdot 1 = 25$	25	0
$(7, 7, 6)$	$7 \cdot 7 \cdot 1 = 49$	28	21
Total	129	108	21

The transport in the last row is justified in Section 6. None of the 21 transported cases is counted as a separately checked refutation.

5 The Boolean encoding

On the labelled vertex set $\{0, \dots, n-1\}$, use a variable x_{uv} for each ordered pair $u \neq v$, meaning that the arc $u \rightarrow v$ is present. The base formula contains the following clauses:

$$\neg x_{uv} \vee \neg x_{vu} \quad (u < v), \quad (1)$$

$$\bigvee_{u,v \in T, u \neq v} x_{uv} \quad (|T| = 3), \quad (2)$$

$$\bigvee_{1 \leq i < j \leq 4} \neg x_{v_i v_j} \quad ((v_1, v_2, v_3, v_4) \text{ distinct}). \quad (3)$$

Thus all 24 orders of each four-set are included. The first family forbids digons, the second forbids independent triples, and the third forbids precisely complete transitive four-sets. Conversely, every $\{I_3, L_4\}$ -free graph satisfies these clauses. The local directed coverage formulas use the analogous three-vertex ordered clauses to forbid L_3 ; tournament formulas additionally require $x_{uv} \vee x_{vu}$. Root and internal-template requirements are unit clauses fixing both arc variables for each prescribed unordered pair.

5.1 Degree constraints

The branch formulas also impose the proved degree restrictions. For input literals $\ell_1, \dots, \ell_m$, a sequential threshold variable $s_{i,j}$ is forced whenever at least j of the first i literals hold. For the applicable indices, use the implications

$$\ell_i \Rightarrow s_{i,1}, \quad s_{i-1,j} \Rightarrow s_{i,j}, \quad (s_{i-1,j} \wedge \ell_i) \Rightarrow s_{i,j+1}.$$

Forbid $s_{m,k+1}$ to impose an at-most- k bound. Induction shows that any actual threshold forces its variable, so this excludes every violating assignment. Conversely, setting variables to their actual threshold truth values extends every satisfying input assignment. The one-way implications therefore suffice; an equivalence at every auxiliary variable is unnecessary. Trivial bounds and empty inputs are handled by their literal cardinality meanings.

Outdegrees and indegrees count their respective arc variables. Total degree counts both, which is valid because (1) excludes antiparallel arcs. A nonedge variable z_{uv} has clauses

$$(\neg z_{uv} \vee \neg x_{uv}), \quad (\neg z_{uv} \vee \neg x_{vu}), \quad (z_{uv} \vee x_{uv} \vee x_{vu});$$

these give exactly $z_{uv} \leftrightarrow (\neg x_{uv} \wedge \neg x_{vu})$. Inside a directed root block the directed degrees are at most two by Lemma 4's proof. The global directed upper bound is eight, replaced by seven throughout the low-degree branch. Subsequent global lower bounds of six are encoded by at most $20 - 6 = 14$ true complemented arc variables in a directed degree list of length 20. Their logical prerequisites are established next.

6 Exclusion order, symmetry, and conclusion

6.1 Noncircular degree strengthening

The finite refutations are used in the following order. First, the three $(8, 8, 4)$ cases and the 25 $(8, 5, 7)$ cases are refuted using only the base constraints, the elementary degree bounds, and fixed templates. These imply, for any hypothetical avoiding graph on 21 vertices,

$$d^+(u), d^-(u) \geq 6, \quad d^+(u) + d^-(u) \leq 15 \quad \text{for every } u. \quad (4)$$

Indeed, total degree is at least 13. A directed degree at most five, together with the opposite bound eight, forces the profile (5, 8, 7) or (8, 5, 7), excluded up to reversal. Total degree 16 forces (8, 8, 4).

Next, the 21 (8, 7, 5) cases are refuted. One uses only the base constraints and twenty use (4). These exclusions and their reversals prohibit total degree 15. Therefore

$$6 \leq d^+(u), d^-(u), \quad 13 \leq d^+(u) + d^-(u) \leq 14. \quad (5)$$

The remaining 31 (8, 6, 6) cases and the low-degree (7, 7, 6) cases use (5). No refutation used to derive a restriction assumes that restriction in its own formula. The independent audit checks the recipe selected for each actual certificate, not merely this intended dependency ordering. Earlier computations on orders 22–24 are not prerequisites of this chain.

6.2 Block automorphisms

In the remaining formulas, each permutation used for symmetry reduction is checked to preserve every directed pair of its fixed block template. Extend it by the identity outside that block. It preserves the root, templates, forbidden subgraphs, and global degree constraints. The product of the three block automorphism groups acts on graph assignments. Order the entire directed-arc word lexicographically, with $0 < 1$. A finite orbit has a minimum. Such a minimum is no greater than its image under every individual block automorphism; consequently these comparisons preserve at least one member of every orbit. Enumerating every element of the product group is unnecessary.

For words a, c , a comparison $a \leq_{\text{lex}} c$ is encoded using equality-prefix variables. Start with a true prefix p . For the current bits a_i, c_i , impose

$$(\neg p \vee \neg a_i \vee c_i), \quad (\neg p \vee a_i \vee c_i \vee q), \quad (\neg p \vee \neg a_i \vee \neg c_i \vee q),$$

where q is the next prefix variable. Equality under a true prefix forces the next prefix. Thus the first unequal bits cannot be (1, 0). Conversely the actual equality-prefix values satisfy the clauses whenever $a \leq_{\text{lex}} c$. Identical-variable comparisons may be omitted. Auxiliary degree and prefix variables can be assigned afresh after choosing the minimum graph; the argument does not assume that these auxiliary variables transform like arcs.

6.3 Reversal transport

The seven order-seven directed templates and the order-six tournament template used in the last branch are all self-converse. Explicit permutations witnessing this are stored and checked on every ordered pair. Global reversal interchanges the root's two directed blocks. Relabelling with the checked permutations restores the templates, so a case indexed by directed classes (i, j) is carried to (j, i) . The non-neighbour template is restored as well. Forbidden configurations and all degree constraints are invariant under this operation. There are $7 \cdot 8 / 2 = 28$ unordered pairs with repetition, covering the 49 ordered pairs. Each transported graph can then be lex-minimized within its new block orbit; the two symmetry-broken formulas need not be literally identical under transport.

6.4 Certified finite exclusions

For each selected formula that was simplified, the simplifier stores a chronological reason clause for every forced literal. Replay checks that all other literals of that original clause have already been falsified. The remaining formula is reconstructed and compared byte for byte with the solver input. A recorded conflict must likewise be a fully falsified original clause. These steps justify the passage to the residual formula, independently of the simplifier's assertion that it is valid.

The eight classification formulas and the 108 selected branch formulas have LRAT refutations. The stricter checker described below validates each proof against its exact input formula and requires a validly derived empty clause. The completed independent replay reports eight checked classification proofs, 108 checked branch proofs, and 129 covered cases. This is an exact Boolean certificate computation; the values are not extrapolated from small tests or inferred from solver timeouts.

Proof of Theorem 2. Suppose an avoiding graph on 21 vertices exists. By Lemma 5, choose a root in one of the five branches, using global reversal if necessary. The certified local cover relabels its three blocks into one of the 129 cases. Apply the exclusions in their stated order. The prerequisite cases are impossible by their direct refutations. In the remaining cases the graph satisfies the justified degree refinements and extends to all auxiliary variables. If needed, transport it to one of the 28 low-degree representatives and choose a lexicographic minimum in its block orbit. It then satisfies a formula whose exact refutation has been checked, a contradiction. This proves $k(3,4) \leq 21$. Proposition 3 proves the reverse inequality. Every graph on more than 21 vertices also contains one of the forbidden configurations by restricting to any 21 vertices. $\square$

7 Reproducibility and the checker correction

The accompanying research archive retains the exact graph input, local representatives and normalized records, coverage formulas, branch formulas, reduction traces, refutations, verifier sources and licenses, explicit reversal permutations, and the independent audit results. The companion `REPRODUCIBILITY.md` gives their locations and the commands for the accepted replay. The manuscript alone, or a solver log alone, is not a replacement for this archive. No permanent public archive identifier is asserted here.

Discovery used finite enumeration, SAT search, preprocessing, and proof conversion. These procedures proposed objects and produced candidate refutations. Exact certification checked the resulting mathematical objects and proof steps. The accepted replay requires Python 3.9 or later and a C99 compiler available as `cc`; no SAT solver, network connection, private classification input, or preinstalled LRAT checker is required once the complete archive is available. The stricter driver compiles the deposited checker source in temporary storage and replays all selected proofs without skipping reductions.

The deposited C checker is `lrat-check.c` from DRAT-trim [1]. The exact deposited revision and source location are recorded in the companion reproducibility guide. An adversarial test in the independent audit found a soundness defect in that deposited source. For the satisfiable formula consisting solely of the unit clause x , the following DIMACS input and purported LRAT proof were incorrectly accepted:

CNF:	purported proof:
p cnf 1 1	2 0 0
1 0	

The proof adds an empty clause without supporting hints. The original checker could reach its pure-pivot shortcut with a zero pivot and accept this unsupported empty clause. Its prior success reports consequently cannot justify this theorem.

The audit driver creates a temporary source copy. Immediately after the successful reverse-unit-propagation conflict return in `checkClause`, it adds

```
if (size == 0) return FAILED;
```

Thus an empty clause can pass only after an actual propagation conflict. The driver checks three controls: the unsupported empty clause is rejected, an empty clause with a nonconflicting single-unit hint is rejected, and the empty clause derived from contradictory units is accepted. It then replays the entire selected corpus with this stricter checker. All eight classification and 108 branch proofs pass. The accepted proof relies on this corrected replay, not on the original checker or an assumption that the defect happened to be irrelevant.

The independent audit also reviewed propagation, resolution-asymmetric tautology coverage of opposite-pivot clauses, deletion handling, and the final empty-clause requirement. The empty-clause correction and controls do not constitute formal verification of all checker code. The trusted base still includes the mathematical encoding and coverage argument, finite reconstruction and isomorphism routines, the corrected C checker, Python, the C compiler, and the execution platform. Branch reduction replay shares recipe code with formula construction; the audit therefore separately inspected the semantics of those recipes. Only the small classification formulas have the separate definition-level clause reconstruction described above. We do not claim an independent second encoding of the complete upper-bound problem.

8 Limitations and assistance disclosure

The result concerns exactly the stated oriented-graph parameter. It does not address digraphs with antiparallel arcs, incomplete acyclic patterns in place of tournaments, other Ramsey parameters, or a classification of all extremal graphs on 20 vertices. The upper-bound cover is unrestricted within the stated domain; the cyclic form of the lower witness is not an assumption on upper-bound counterexamples. No asymptotic, probabilistic, or floating-point inference occurs in the accepted proof.

The literature comparison is bounded to the cited sources, inspected on 5 September 2026. It does not certify exhaustive novelty or continued open status. The earlier claim of the same endpoint remains separate from the verification performed here. Neither a novel SAT symmetry method nor uniqueness of the lower-bound example is asserted.

Computational and language-model assistance. Language-model agents were used extensively in the research workflow: for developing and criticizing mathematical reductions, writing and inspecting programs, organizing exact computations, examining sources, and drafting this manuscript. A separate fresh-context language-model agent performed the recorded independent computational audit, including its independently implemented witness test, checker negative controls, and stricter certificate replay. Here “independent audit” denotes that separation from the discovery context and those concrete checks; it is not a claim of external human peer review. The present manuscript was prepared by a language-model agent for release preparation using that accepted audit. The audit’s acceptance does not establish literature priority or certify the prose of this manuscript. Language-model confidence is not evidence for the theorem; the evidence is the explicit reductions, exact witness, coverage checks, and corrected refutation replay. No proof-assistant formalization or verification of compiled machine-code semantics is claimed.

References

- [1] The DRAT-trim proof checker. Software repository, <https://github.com/marijnheule/drat-trim>. File `lrat-check.c`; exact deposited revision documented in the companion reproducibility guide; accessed 5 September 2026.

- [2] Erdős problem 112 — live audit, literature correction, exact small cases, and a cyclic-Cayley census. Working report, Erdős Problem a Day, <https://erdosproblemaday.com/report/112>, 2026. Dated 12 August 2026; Section 6. Accessed 5 September 2026.
- [3] Ferdinand Ihringer, Deepak Rajendraprasad, and Thilo V. Weinert. New bounds on the Ramsey number $r(I_m, L_n)$. arXiv:1707.09556v3, 2020. Version 3, revised 8 April 2020.
- [4] Daniel Kirtchakov. Certify: replayable certificates for machine-checked mathematics. Software repository, README, Part G, <https://github.com/05oz/certify/blob/main/README.md>, 2026. Repository account 05oz; accessed 5 September 2026.
- [5] Zijian Zeng. One-sided parity compression for certified SAT attacks on the 33-point Erdős–Szekeres problem. SSRN preprint, 2026.
- [6] Zijian Zeng. The small Ramsey degree of an edge in ordered split graphs is three. Preprints.org, August 2026. Version 1, posted 31 August 2026; not peer reviewed.